



Figure 1: Haxe components

- Haxe is available for download with an open source licence.
- It can be configured in various operating systems without any major glitches.
- The Haxe programming language is fully featured.
- It has the capability to target both programming languages and virtual machines like Neko.
- Haxe has support for modern programming languages like C#, Python, etc.
- It can be used to compile into client side scripts like JavaScript and server side scripts like PHP.

The evolution of Haxe

Development on the Haxe project was started in 2005 by Nicolas Cannasse. Initially, it was named as 'haXe'. The milestones in Haxe's development are listed below:

- The first beta version was released in February 2006. The targets it supported then were AVM bytecode and Neko, which was the virtual machine developed by Cannasse himself.
- The version 1.0 of Haxe was released during May 2006. It had support for JavaScript.
- In August 2006, the 'haxelib' tool was added as a component to the Haxe Toolkit.
- ActionScript 3 was added as a target in March 2007.
- In July 2008, Haxe version 2.0 was released with support for PHP.
- In July 2009, C++ support was added to Haxe with the release 2.04.
- In January 2011, version 2.07 was launched, which supported 'macros'.
- The versions 2.08 and 2.09 focused on the improvements in the JavaScript target.
- In 2012, support for Java and C# was added in Haxe 2.10.
- Haxe 3 was launched in May 2013 with the establishment of the Haxe Foundation to deal with future

developments and maintenance.

- The latest version of Haxe is 3.20, which was released this year.

Haxe installation

The latest Haxe version can be downloaded from <http://haxe.org/download/> which provides both the installer and binaries for Windows, Linux and Mac based computers. A typical Haxe installation on a Windows PC is easy (similar is the case with other OSs like Linux). Just download the executable installer file and run it using a double click. Being a small file, the installation will get completed in a few minutes. During the installation, you will be given an option to select or remove the Neko virtual machine too.

The Haxe cross-compiler

The Haxe compiler is capable of compiling the Haxe source code into the source code of the target language. The available programming language targets for the Haxe compiler are given in Table 1.

Language	Potential applications	Available since
JavaScript	Web, desktop, API	Beta (2006)
ActionScript 3	Games, mobile, API	1.12 (2007)
PHP	Web	2.0 (2008)
C++	Games, CLI, mobile, desktop	2.04 (2009)
Java	CLI, mobile, desktop	2.10 (2012)
C#	Mobile, desktop	2.10 (2012)
Python	CLI, Web, desktop	3.2 (2015)

Table 1: Haxe's programming language targets

Apart from the above specified source-to-source compilation, Haxe supports source-to-bytecode compilation as well, as shown in Table 2.

Name	Potential application types	Available since
Flash	Games, mobile	Alpha (2005)
Neko	Web, CLI	Alpha (2005)

Table 2: Haxe virtual machine targets

It is interesting to know that Neko has both programming and virtual machine components.

The Haxe 'Hello World'

The syntax of the Haxe programming language would look very familiar to most programmers as it is more oriented



Figure 2: The evolution of Haxe